18.5.4. Radiation therapy with internal radionuclides

Radiotracers are also used for therapy though to less extent than in diagnosis (C3b in Table 18.3). The main application is the use of ^{131}I for treatment of thyreotoxicos (Graves disease with enlargement of the thyroid gland), thereby reducing the function of the thyroid. Some data for ^{131}I use: amount administered ca. 200−1000 MBq, organ dose to the thyroid ~340 nGy/Bq (total dose to the glands >500 Sv); whole body dose ~0.04 nGy/Bq. ^{32}P compounds are used in the bloodstream to destroy excess red blood cells in polycythemia. Diphosphonate compounds localize in normal bone tissue, but concentrate in cancerous bone at relatively higher levels (5−15 times higher). Patients receiving this type of radiopharmaceutical experience swift lowering of pain levels.

One of the most fascinating aspects of using radionuclide labelled compounds in medical therapy is to develop monoclonal antibodies, which will seek out particular types of cancer cells and bind to them. If a β-, or α-emitting radionuclide is attached to these antibodies, they will deliver a large absorbed dose to the cancer cells without needlessly irradiating the surrounding tissues. Much research is directed towards this goal of cancer therapy.

18.6. INDUSTRIAL USES OF RADIOTRACERS

Industry has applied radiotracers in a very large variety of ways. More than half of the 500 largest manufacturing concerns in the United States use radioisotopes in the production of metals, chemicals, plastics, pharmaceuticals, paper, rubber, clay and glass products, food, tobacco, textiles, and many other products. Radioisotopes are used to study mixing efficiency, effect of chamber geometry, residence time in reactors, flow rates and patterns in columns and towers for fractionation, absorption, racemization, etc. Some of the many uses are listed in Table 18.6 and a few are described below to reflect the scope and value of the industrial applications of radioisotopes. Quite often the radionuclide used is not isotopic with the system studied.

18.6.1. Mixing

Mixing is an important mechanical operation in many industries. Poor mixing may give an unsatisfactory product and low yield of the operation; unnecessary mixing is a waste of time and energy. By adding a radionuclide to the mixing vessel, or by labelling one of the components, the approach to mixing equilibrium can be followed either by external measurement or taking samples at different time intervals. Among examples of this technique are fluxing of cement, gravel, sand, and water to concrete measured by using irradiated pebbles. The homogeneity of glass melts can be determined by adding $^{24}NaHCO_3$ to the melt; organic compounds of ^{95}Zr ($\beta^-\gamma$ $t_{½}$ 64.02 d) have been used to follow homogenization of oil products. Other uses have been the addition of vitamins to flour, coal powder to rubber, water and gas to oil fields, etc.

Table 18.6 Radioisotope technique studies at ICI[†] undertaken in a typical year

Technique	Number of applications
Level and interface measurements:	
Gamma-ray absorption	210
Neutron backscatter	480
Gamma-ray backscatter	71
Blockage detection and deposition:	
Gamma-ray absorption	132
Neutron backscatter	129
Entrainment and voidage:	
Gamma-ray absorption	86
Thickness and corrosion measurements	15
Distillation-column scans	108
Flow measurements:	
Pulse velocity	483
Dilution techniques	84
Leak detection	90
Residence-time studies	21
Carryover studies (tracer)	6

The list is not fully comprehensive as less commonly used techniques have not been included.
[†]Imperial Chemical Industries (UK).

18.6.2. Liquid volumes and flows

The liquid volume of a closed system may be difficult to calculate from external container dimensions, particularly if there is a mixing action, either by external circulation or by internal stirring. For example, sulfuric acid volume is desired in an alkylation plant where extensive intermixing between acid and hydrocarbon prevents a well-defined level from forming. ^{137}Cs is added to the sulfuric acid and from the dilution of the added tracer the total volume of acid is calculated.

Another method is applicable to determining volumes of tanks through which there is a known constant flow. A tracer batch is put into the incoming line. Mathematical treatment, assuming complete immediate mixing of the incoming stream with the vessel contents, predicts that the tracer concentration in the tank falls off exponentially with a rate determined by throughput F and volume V according to the equation

$$R = R_0 \, e^{-Ft/V} \tag{18.37}$$

where R_0 is the counting rate at zero time (break through of radioactivity at tank exit) and t is the time when the count rate R is obtained at the same point.

The flow rate F of rivers and streams can be measured by injection of a radionuclide and measurement of the time for its arrival at detectors placed downstream. Because of

turbulence the radioactive "cloud" becomes quite diffuse. Therefore a more efficient technique called "total count" is used. A known amount (A_0) of the radionuclide is injected into the river and a downstream detector registers a total count (R_{tot}) as the radioactivity passes; the faster it passes, the lower is the measured radioactivity. Thus

$$F = \psi\, A_0 / R_{tot} \tag{18.38}$$

ψ is the counting efficiency which has to be determined under known conditions. The technique takes into consideration both longitudinal and transversal mixing.

18.6.3. Wear and corrosion

Wear and materials transfer are easily followed if the material undergoing wear is made radioactive. This has been used for studying wear of parts in automobile engines, cutting tools, ball-bearings, furnace linings, paint abrasions, etc. In this case it is important that the surface undergoing wear has a high specific radioactivity. If the material cannot be tagged by adding (e.g. plating) a radionuclide, the material has to be activated by irradiation (e.g. by accelerator) to produce the radioactive species.

Corrosion in gas and oil pipe-lines on the sea floor is monitored by welding patches containing a series of long-lived radionuclides at different material depth into the pipe wall during construction of the pipe-line. Wall corrosion will remove one isotope after the other as it proceeds. The presence of these isotopes can be monitored from time to time by sending a capsule with a suitable detector and recorder through the pipe.

18.6.4. Chemical processing

We have described some areas of application of radiotracers which are useful to the chemical industry. To be more general, radioactive tracers are used to seek information of flow and mixing patterns for

- parameter evaluations (mass balance, flow rates, kinetics)
- fluid dynamics (trouble shooting, residence time distributions, modelling flow phenomena)
- chemical reaction engineering (effect of flow and mixing on conversions, modelling of reaction systems, process control algorithms).

Each application is rather case specific and described in the chemical engineering literature.

18.7. ENVIRONMENTAL APPLICATIONS

Nuclear weapons tests have released large amounts of radionuclides into the atmosphere, which through their own weight or by rain have been carried to the earth's surface. Geophysics has made use of this weapon "fallout". By measurements on T (as HTO